

UNIT - I

ENVIRONMENT:

sum total of all social, economical, biological, physical & chemical factors that surround, effect & influence the survival, growth & development of humans, plants and animals who are both creators & modulators of the environment.

ENVIRONMENTAL STUDIES:

systematic understanding of our environment and our existence in it. It includes many interconnected problems involving human population, natural resources & environmental problems.

OBJECTIVES OF EVS:

(2 guiding principles)

- ① Awareness: to acquire awareness & sensitivity to the total environment & its allied problems.
- ② Knowledge: to gain experience & acquire basic understanding of the environment & its associated problems.
- ③ Attitude: to acquire set of values & develop feeling of concern for the environment & ~~its associated problems~~ build motivation for active participation in environmental improvement & protection.

- (4) Skill : to develop skills in identifying & solving environmental problems.
- (5) Participation : to provide an opportunity to be actively involved at all levels in working towards solving environmental problems.

SCOPE OF EVS.

- wide & related to every field.
- Ecosystem structure & function
- Research & development
- Natural Resource conservation
- Environmental pollution control
- Environmental management
- Industry
- Social development
- Environmental journalism
- Environmentalist
- Green advocacy & marketing.

IMPORTANCE OF EVS

1. provides us basic knowledge of environment
2. helps in understanding current environmental problems
3. helps in conservation & management of natural resources
4. helps in maintaining ecological balance and controlling environmental pollution.
5. helps in sustainable development.
6. helps in educating people regarding their duties towards the environment.
7. helpful in environmental management
8. aims to protect biodiversity
9. studies the agricultural & design of sustainable production.
10. gives information about related to population explosion, growth & development, impact of population growth on the natural resources & national economy
11. helps in developing social responsibilities towards protection of environment.

COMPONENTS OF ENVIRONMENT:

1. Biotic :

living components

eg: microbes, plants, animals, human beings

2. Abiotic

non-living / physical components

eg: air, water, soil, rocks etc

ECOSYSTEM:

→ The natural unit, consisting of living & non-living part which interacts to form a stable system.

→ It is an area within the natural environment & possesses all characteristics required to sustain life

→ Eg: Pond, lakes, forest, grassland etc

→ Balanced ecosystem -

nutrients are able to cycle efficiently & no community or any natural phenomenon is interrupting the flow of energy and nutrients to other parts

STRUCTURE OF ECOSYSTEM.

1) Abiotic components :

all non-living components of environment

• inorganic → involved in mineral cycle

Eg: C, S, N, P, K

- Organic substances - form the living body & influences the functioning of ecosystem
Eg: Lipid, Protein, Carbohydrate
- Climatic factor: air, temp, soil, water, sunlight.

2) Biotic components:
all living components of the environment.

a) Autotrophs / Producers:

these organisms produce their own food as they contain chlorophyll & capable of directly obtaining energy from sun & convert, store it in chemical energy in their bodies.

① Photoautotrophs - fix energy from the sun.

Eg: Green plants

② Chemo-Autotrophs - bacteria which oxidise reduced inorganic substances such as NH_3 , Sulfur to complex compounds.

Eg: Nitrifying bacteria

b) Heterotrophs / Consumers:

obtain food from by consuming producers.

① Herbivores : depend on plants for food & energy.

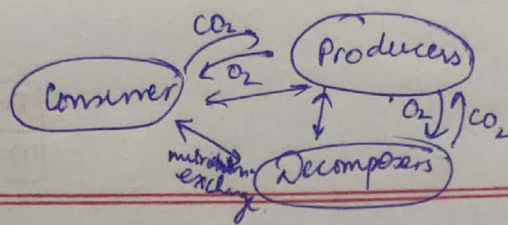
Eg: insect, rat, goat, deer etc.

② Primary Carnivore: depend on ~~for~~ herbivores for food & energy

Eg: frog, bird, snake, cat etc.

③ Secondary Carnivore: depend on primary carnivores.

Eg: Tiger, lion



c) Decomposers / Saprotrophs:

consuming dead organisms, plants, animals & decomposing producers & consumers into simple compounds.
Eg. Bacteria, Fungi

FUNCTION / IMPORTANCE OF ECOSYSTEM:

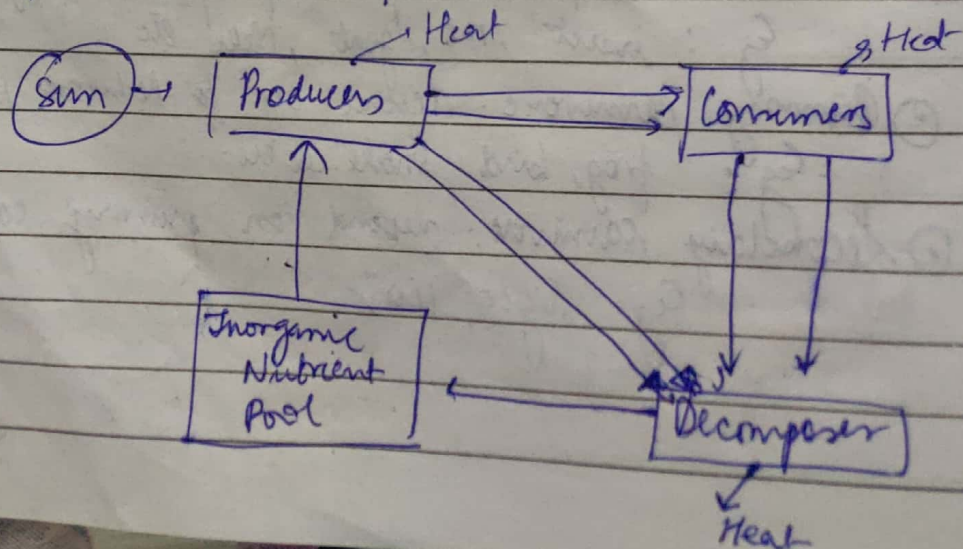
1) Ecological succession

- It is an orderly change in the composition or structure of an ecological community over time.
- A community begins with relatively a few living organisms & becomes complex over a period of time & hence become stable.

Characteristics:

- increase productivity
- shift of nutrients from reservoir
- increase diversity of organisms which increase niche development.
- Gradual increase in complexity of food webs.

ENERGY FLOW THROUGH ECOSYSTEM.



detritus → dead organic matter

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- Sun is the major source of energy
- In food chain, energy flows from one organism to other when one organism eats another ~~and~~ organisms.
- Nutrients can be cycled through an ecosystem but energy is simply lost over time
- In ecology, energy flow (or caloric flow) refers to the flow of energy through a food chain
- Solar energy not used in an ecosystem is eventually lost as heat
- Inorganic nutrients are recycled, the energy is not.
↳ explain flowchart.

→ TROPHIC level :

- representation of energy flow in an ecosystem
- It is the position it occupies in a food chain

FOOD-CHAIN:

- The transfer of food energy from producers to consumers through a series of trophic level
- Each step in food chain is called trophic level
- A food chain starts with producers & ends with top carnivores.

Eg: Grasses → Grasshopper → Frog → Snake → Eagle

- At each linkage in the chain, a major part of energy from food is lost for daily activities.

④ Detritus food chain → starts from dead organic matter of decaying animals & plant bodies. Dead organic matter are called detritus/decomposer. They are eaten by predators

Dead plants - soil microbes → insects → lizard

FOOD WEB:

- The interlocking pattern of various food chains in the ecosystem.
- Many food chains are connected & overlapped because most consumers feed on multiple species.
- It is a graphical representation of what eats what in an ecological community.
- If due to any human activity, any link in the web is broken, then the species linked with the web may become extinct.

→ Energy flow in food web:

Grass → insects → fishes → birds → tigers.

Grass → deer → tigers.

Grass → rats → eagles → tigers.

- The food web provides more than 1 alternative for food to most organisms in an ecosystem & therefore increases their chance of survival.

ECOLOGICAL PYRAMID:

- Graphical representation of structure & function of trophic level of an ecosystem in the form of pyramid.
- In ecological pyramids, producers form the base & top carnivores form top of the pyramid. Other consumer trophic levels are in between.

always start from base.

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→ The pyramid consists of no. of horizontal bars depicting specific trophic levels.

→ The length of each bar represents the total no. of individuals or biomass or energy at each trophic level in an ecosystem.

→ Ecological pyramids:

a) Pyramid of numbers

b) Pyramid of biomass

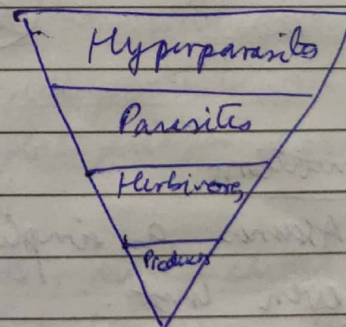
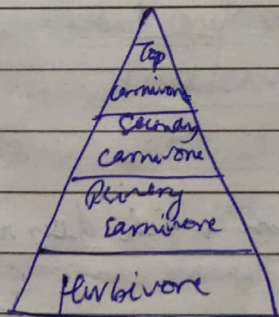
c) Pyramid of energy

a) Pyramid of No → represents total no. of individuals of different species at each trophic level.

It may be upright or inverted in nature depending on size of ecosystem.

Eg: Upright → Grassland ecosystem

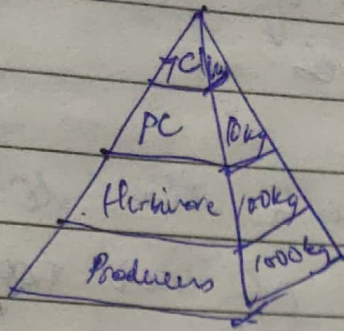
Inverted → Tree ecosystem.



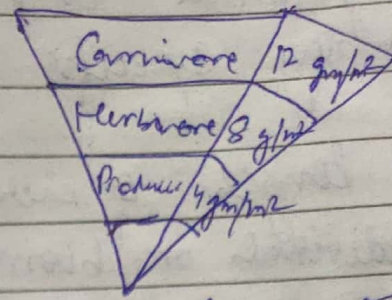
b) Pyramid of biomass.

↳ by collecting all organisms occupying each trophic level separately & measuring their dry weight.

↳ represents total biomass available per unit area in each trophic level. May be upright or inverted in nature.

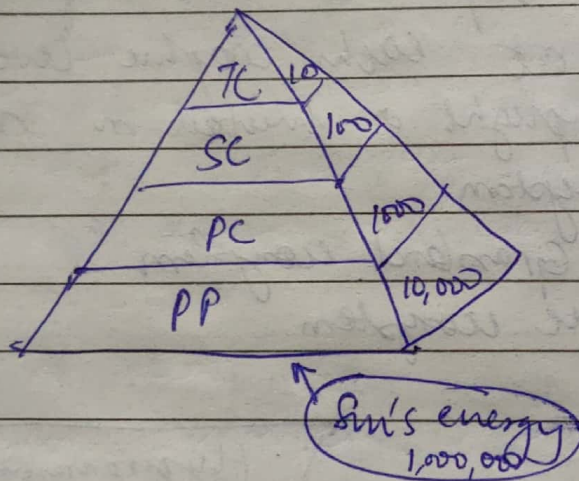


any ecosystem on land.



Aquatic system

- 1) Pyramid of energy represents amount of energy at each transfer to another trophic level. Always upright, with large energy loss.



Limitations:

- Assumes a single food chain which does not even exist.
- Saprophytes not given any place.
- does not consider that same species belonging to two or more trophic level.

intro, types, features, structure, func:

forest, grassland, desert

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(pond, rivers, lakes, river, oceans, estuaries)

FOREST ECOSYSTEM:

- supports innumerable plant & animal species as well as ground cover, water & air sub-systems.
- Each forest type forms a habitat for a specific community of animals that are adapted to live in it.

VERTICAL STRUCTURE - distinct layers, each adapted to increasingly filtered sunlight if going top down.
canopy, understorey, ground layer & forest floor.

HORIZONTAL STRUCTURE: composition & distribution varies depending on moisture, drainage, slope, slope aspect, soil type & light intensity.

ENVIRONMENTAL STRUCTURE:

soil type, temp, moisture, precipitation pattern etc

CLASSIFICATION:

- ① Himalayan coniferous - pine, deodar, wild goat & sheep
- ② Himalayan broadleaved - maple, oak, Himalayan tahr, leopard
- ③ Evergreen north east, western ghats, Andaman, Nicobar → Javan, ficus, tiger, sambar
- ④ Deciduous - dry - teak, aml, tiger, chital
- ⑤ Thorns & scrub, semi-arid - babul, ber, neem, blackbuck, monitor lizard
- ⑥ Mangrove Delta - avicennia, coriaria, fish.

GRASSLAND ECOSYSTEM

- vegetation dominated by grass & non woody plants.
- have limited rainfall

→ Eg: floodplains of rivers coming out of Himalayas onto Indo-Gangetic Plain - mainly consists alluvial sediments b/w Yamuna & Terai. Deer grassland - one horned rhino, asian elephants, sambar deer, bharal tiger.

DESERT ECOSYSTEM:

→ barren area of land with scanty rainfall
& hostile living conditions for plant & animal life.

eg: cold desert → Ladakh

frequent dust storms → sand dunes.

→ sparse grass, shrubs → which grow if it rains.

Summer → Great & Little Rann of Kutch → arid ecosystem.
Salt marshes Wild an only breeding place of flamingo
Noon → Indian wolf, desert cat, desert fox,
attractive aquatic birds

AQUATIC ECOSYSTEM:

→ ecosystem of waterbody

→ plants & animals that can survive in water environment.

a) Marine / Ocean.

→ contain dissolved compounds specially salts giving salinity.

→ Many zones depends upon: depth & shoreline features:

Oceanic zone: whales, sharks, tuna

Shore zones → salt marshes, coral reefs, lagoons.

b) Estuaries.

→ fresh water + salt water.

→ found on coast (river + sea)

→ most fertile

→ Organisms tolerant to physical changes.

c) Lake

- completely surrounded by land
- either fresh or saline water.
- Algae, herbivores fish, snails.
- not permanent = water can sink or evaporate

d) Streams / rivers

- flowing water ecosystem
- zones determined by river bed gradient or velocity of current.

e) Ponds

- small freshwater bodies (1 m^2)
- shallow & still water
- Algae, snails, beetles, water bugs.

BIODIVERSITY:

- It is the variety & variability among all groups of living organisms & ecosystem in which they occur.
- Refers to study of all living organisms.

LEVELS:

1) Genetic diversity

- variation in genes with same species.
- Genes are information packet of living cells.
- It allows species to adapt to changing environments.

Eg: 1000 diff varieties of mangoes.
Alphonso, Langra, Dasheri, Kesar

- 2) Species biodiversity:
- variety & variability among species within a given ecosystem.
 - Eg: tropical region near ecosystem.
 - It incorporates both: species richness and evenness of species.
no. of species
proportion of species at a site

- 3) Ecosystem biodiversity:
- variation in structure & function of ecosystem
 - tells about food & energy flow & stability of ecosystem.
 - Eg: River ecosystem, forest ecosystem
 - Most complex level of diversity:

VALUE OF BIODIVERSITY:

Direct value:

- a) Consumptive use value:
- direct utilization of timber, food, fuel, wood by local communities.
 - daily needs, food, building material.
 - plants, fish, fuel, drugs & medicines.
- b) Productive use value:
- product is commercially sold in national & international markets
 - Eg: Textile, leather, silk, paper.
 - traded in market & fetch a booming business.

INDIRECT VALUES:

a) Social & cultural values:

- plants & animals considered holy & sacred & worshipped
- Tulsi, Peepal, Cow, snake etc.
- Cultural value is given to forest & peacock & lotus are national bird & flower.

b) Ethical:

- values related to conservation of biodiversity
- all life forms must be preserved
- moral right to exist independent of our need of them.

c) Aesthetic value:

- natural landscape at undisturbed places → delight to watch
- provide opportunities for recreational activities like birdwatching, photography etc.
- promotes eco-tourism which also generates revenue.

a) Option values:

- unexplored or unknown potential of biodiversity
- keeping future possibility open for their use.

BIODIVERSITY AT GLOBAL, LOCAL AND NATIONAL LEVELS:

a) Global:

- 8.7 million different forms of life.
- Majority of species are yet to be discovered.

- Most bio-rich nations in south which are developing nations.

~~NATION~~

b) National:

- India has rich biological diversity of flora & fauna.
- Wide spectrum of habitats from tropical rainforest to alpine vegetation.
- 5 world heritage sites & 12 bio-reserves.

INDIA AS A MEGA DIVERSITY NATION:

- rich at all levels:
- represents almost all biogeographic regions
- covers wide spectrum of habitats.
- 12th among list of mega diversity nations.

BIO-GEOGRAPHICAL CLASSIFICATION OF INDIA:

- 10 major regions depending on geography, climate & pattern of vegetation & communities of mammals, birds etc.
 - Each contains variety of ecosystem.
- Bio-geographic zones.
- (1) TRANS-Himalayan region of Ladakh covering mountains with cold snow
 - (2) Himalayan ranges & valleys of Kashmir, Uttarakhand & north east
 - (3) The Terai, the lowland where Himalayan rivers flow into plains.

- ① The gangetic & Brahmaputra plains
- ② The Thar desert of Rajasthan
- ③ Semi-arid grassland region of Deccan Plateau.
Gujarat, Maharashtra, Karnataka.
- ④ Northeast states of India
- ⑤ Western Ghats in Maharashtra, Karnataka, Kerala
- ⑥ Andaman & Nicobar islands
- ⑦ Long western & eastern coast belt with ~~land~~
sandy beaches, forest & mangroves.

HOTSPOTS OF BIODIVERSITY:

- areas on earth which exhibit high species richness as well as high species endemism.
- It must contain 0.5% of global plant species as endemic
- must have occupied 70% of its land with plant & animal species.

→ 34 hotspots in the world

→ Countries which have relatively large proportion of these hotspots are megadiversity regions.

→ India has 2 hotspots:

a) Western Ghats:

- A major portion of amphibian & reptile species, especially snakes are concentrated.
- habitat of 1500+ endemic plant species

b) Eastern Himalayas:

- contains 30% of the endemic species.

RET SPECIES:

- ① Rare species:
 - which are already less in no. & need to be preserved.
 - neither endangered nor vulnerable, but are at risk.
 - eg: Hornbill, Himalayan Brown bear
- ② Threatened species:
 - considerably becoming less in no. due to natural reason or human activity.
 - eg: White tiger, Rhino, Sparrow.
- ③ Endangered:
 - when no. has been reduced to critical level
 - if not protected, immediate danger of extinction.
- ④ Vulnerable:
 - likely to become endangered
- ⑤ Extinct:
 - not been observed in last 10-15 yrs
 - eg: Dodo, Dinosaurs.
- ⑥ Endemic species:
 - habitat restricted to particular area.
 - often endangered

THREATS TO BIODIVERSITY:

1) Habitat loss:

→ cause: urbanization, deforestation

→ Approx 18 million acres of forest are lost each year.

2) Poaching of Wildlife:

→ illegal killing of wildlife for sale in trade market

→ skin & bones of tigers, ivory of elephant, horns from rhinos & perfume from musk deer

→ corals & shells → for export or sold on beaches

3) Man-wildlife conflict:

→ interaction b/w wild animals & people

→ negative impact on resources or people or wildlife or habitat

→ happens when growing human population overlap with established wildlife territory, creating reduction of resources/life to people/animals.

BIODIVERSITY CONSERVATION:

1) In-situ conservation (within habitats)

→ most appropriate method

→ find region of high biodiversity & convert it to National park/ Sanctuary etc.

→ long term protection, better opportunity for conservation & evolution.

2) Ex-situ (outside habitat)

→ endangered species close to extinction

→ driven outside natural habitat for carefully controlled situation

- expertise to multiply species in artificially managed conditions..
- forming gene banks: seed banks,
- forming zoo & botanical garden.
- aquaria.
- captive breeding of animals
- Artificial propagation of plants.
- long term protection, more offspring than usual, hybrid species.